

Voice Card & Framed Case Studies (FL-02)

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1. Voice Card (Standing Instruction)

Voice Card: Direct, plainspoken, honest, technical, zero buzzwords.

Standing Instruction for Claude Project:

"Write like a real software and machine learning engineer. Be direct, concise, and plainspoken. Never use buzzwords or filler like 'spearheaded', 'game-changer', 'results-driven', 'delve', or 'seamless'. State what was built, the concrete decisions made, and the exact numbers without inflating them. Always use active voice."

2. Before & After: Generic AI Copy vs. My Edited Version

Version	Copy	Why It Changed
Before (Generic AI)	"I spearheaded an innovative, cutting-edge machine learning solution leveraging state-of-the-art predictive algorithms on massive big data to seamlessly revolutionize organic search optimization and drive unparalleled traffic growth."	Puffed up with meaningless buzzwords ('spearheaded', 'cutting-edge', 'leveraging', 'seamlessly'). Hides what the model actually does.
After (My Edited Voice)	"I built a logistic regression pipeline in DuckDB and scikit-learn that scores decaying search pages across 9.8 million daily rows. Evaluated on unseen client domains, it reached 0.5000 Precision@50, beating FlyRank's fixed heuristic baseline of 0.4200."	Direct, grounded, and specific. Names exact tools, dataset size, split design, and empirical test result.

3. Framed Case 1: Content Refresh Opportunity Scoring (Lead Project)

Beat 1: The Problem

SEO teams manage thousands of published articles. Over time, search traffic decays, but identifying which pages need a rewrite is usually done with blunt rules (e.g. 'any page older than 6 months'). Teams waste hours rewriting pages that would have recovered on their own or miss dying high-traffic pages until it is too late. They needed a model to rank pages by likelihood of continued decline.

Beat 2: What I Did & Decided

- **Data Grain & Slicing:** Queried 9,841,378 daily performance rows from FlyRank warehouse in DuckDB at content page grain for March 2026.
- **Strict Leakage Prevention:** Isolated feature window (March 1–15) from outcome window (March 16–31). Exposed and removed target-derived leakage trap features that falsely inflated AUC to 0.9991.
- **Client Normalization:** Normalized impressions and clicks within each client to prevent memorizing client traffic scales.
- **Honest Split Choice:** Evaluated on a 75/25 GroupShuffleSplit by client_hash_id to measure generalization to unseen client domains.

Beat 3: What Came of It

- Evaluated on unseen clients, Logistic Regression achieved **0.5000 Precision@50** vs baseline heuristic of **0.4200** (+19% relative precision lift).
- Built an Action Playbook with transparent reason codes (stale_visible_page, low_ctr_visible_page, model_decline_risk) for human editorial decision support.
- Deployed live research paper at <https://ridaeman02.github.io/flyrank-ml-internship/>.

4. Framed Case 2: Leakage-Free Temporal ETL & Contract Verification

Beat 1: The Problem: Future observations quietly leak into training features in temporal data, producing models that look great in notebooks but fail in production.

Beat 2: What I Did & Decided: Built a pre-flight assertion suite (scripts/verify_data_contract.py) that deterministically blocks execution if target columns leak into feature matrices or null bounds fail.

Beat 3: What Came of It: Reduced verification time from 20 minutes of manual inspection to under 1.2 seconds, integrated directly into GitHub Actions CI.

5. Bio & Contact Copy (Pointing to the One Action)

Bio: Computer Science undergraduate & Machine Learning Intern at FlyRank AI. Working on backend data pipelines (Python, DuckDB, SQL) and applied search ranking models. Focused on building honest models that work on messy real-world data.

The One Action: Invite me to a technical interview to discuss how my end-to-end DuckDB and scikit-learn pipeline applies to your data challenges. Contact: ridaeman0002@gmail.com | GitHub: github.com/ridaeman02/flyrank-ml-internship